Evaluating Visual Decision Support: How Does Preference Elicitation Shape Metric Sensitivity?

Lena Cibulski, Tobias Mertz, Evanthia Dimara, Stefan Bruckner

***Abstract*—Supporting decision-making is a central goal in visualization research, yet evaluating how effectively visualizations aid decision tasks remains difficult. Objective metrics for decision quality as a baseline for comparative evaluations are rare, with previous work proposing the consistency between a decision made and self-reported subjective preferences. However, the impact of preference elicitation design on such metrics is not well understood. Our focus is therefore on understanding how elicitation methods shape the sensitivity of decision quality metrics, not on comparing visualization techniques themselves. We report a preregistered study with 548 participants examining how varying the expressiveness of preference elicitation affects the sensitivity of choice consistency metrics when comparing parallel coordinates and tabular visualizations. Our baseline condition replicates a previous study using simple preference elicitation with significantly more participants and confirms its inconclusive findings that suggest a comparable performance of parallel coordinates and tabular visualization for the tested decision task. We further test two additional conditions to investigate if more expressive preference elicitation increases the metric's sensitivity. Our results suggest that more expressive elicitation may improve sensitivity in detecting performance differences, but further increases offer no clear additional benefit in our study. We observed these sensitivity differences even though the experimental setup, with its comparable baseline performances, was not optimized for detecting visualization performance differences. Our results provide initial evidence that elicitation design can affect the informational value of decision-centric visualization studies, informing both the development of metrics and the interpretation of comparative studies. A preprint of this paper, along with the pre-registration, data analysis scripts, and all supplemental materials, is available at https://doi.org/10.17605/OSF.IO/U63ZW.**

***Index Terms*—Decision-making, multivariate, preference elicitation, parallel coordinates, tabular visualization, user study.**

I. Introduction

IDENTIFYING the best item among many alternatives is a common task. As tourists, we aim for the nicest hotel to spend our holiday in. As engineers, we decide which prototype is taken to production. As investors, we choose stocks to hold our financial resources. While the settings of these decisions vary, they are challenging for the same reason: each item is described by multiple, potentially conflicting, attributes. An optimal solution rarely exists. Instead, what is considered *best* depends on the subjective assessment of the decision-maker, who needs to balance the attributes to arrive at a compromise. Pivotal information is often hidden in the data, making visualization a promising tool to help people make good decisions. Evaluating visualizations for their ability to support decision tasks is, however, challenging because decisions are inherently subjective. What one person considers a good outcome may not be desirable for someone else. Where decisions cannot be judged based on a single criterion like profit, any attempt to quantify their quality needs to consider the individual perspective of the decision-maker. Until now, most evaluations of decision support facing such a lack of known truth have relied on subjective ratings (e.g., satisfaction and confidence) rather than an objective metric for decision quality.

One approach to defining an objective metric involves measuring the consistency between a decision and the decision-maker's self-reported preferences for attributes. However, such a metric can only be as precise as the self-report of subjective preferences, which include attribute importance and the desirability of attribute values. Interface design likely affects how clearly people can express the latter. For example, Dimara et al. asked participants to indicate whether they preferred high or low values of the decision attributes [1]. While this may work for attributes like price, it may be less suitable when preferences are non-monotonic, such as preferring three hotel rooms over two or four. In such cases, restricting input to binary directionality can lead to imprecise reports and noisy observations in downstream metrics. In fact, Dimara et al. found their decision quality metric to be *"not sensitive enough to capture differences across conditions"* [1].

Inspired by this finding, we aim to investigate how the sensitivity of a metric can be increased. To this end, we follow the principle of measuring the consistency between self-reported preferences and a decision made but employ different methods to elicit participants' preferences. We contribute a large-scale crowdsourced user study, in which 548 participants use three different interfaces to report their preferences on decision attributes and make casual decisions using two different visualizations. The primary goal is to understand how preference elicitation interfaces with varying levels of expressiveness affect the decision quality metric's sensitivity, i.e., its ability to capture meaningful differences between the visualizations. We aim to learn to what extent more expressive interfaces produce preference specifications with more nuances, which we expect to be a precondition for a more sensitive metric. A secondary goal is to provide a conceptual replication [2] of

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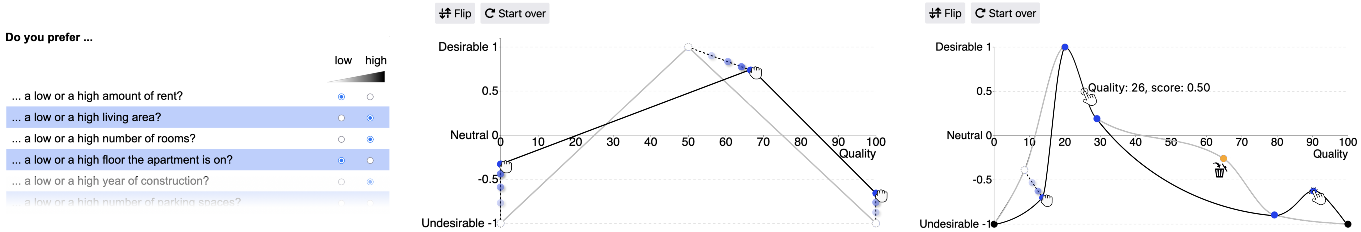


(a) Low Expressiveness (b) Medium Expressiveness (c) High Expressiveness

Fig. 1. We studied the impact of (a) low, (b) medium, and (c) high expressiveness of preference elicitation on the metric sensitivity when assessing the consistency between preference reports and a decision made with parallel coordinates or tabular visualizations. The expressiveness levels offer increasing degrees of freedom for participants to map decision attribute values to desirability scores. Does this lead to equally increasing sensitivity of the resulting decision quality metric or do we observe signs of supersaturation?

previously published research [1] with a substantially larger sample, thereby evaluating its credibility and generalizability. To this end, one of the interfaces in our experiment is chosen to partly reproduce the condition tested in the previous study.

II. Related Work

With our experiment, we study how different preference elicitation interfaces influence the assessment of multi-attribute choices supported by visualizations. Preference elicitation is the *"quantitative expression of subjective judgement"* [3]. With preferences being highly implicit, their elicitation likely comes with a *measurement error*, i.e., a difference between the elicited preferences and the user's unknown true preferences [4]. We review how users can interactively specify decision preferences, and then discuss empirical approaches from visualization research to evaluate decision support quality.

Collecting Attribute-Based Decision Preferences – Information about what people find desirable or undesirable is commonly collected for use in recommender systems [5], multi-attribute ranking systems [6], [7], or visual explorations [8], [9]. Prior work on graphical elicitation emphasizes the role of visual interfaces for externalizing value and beliefs as a source of information for decision-making [10], [11]. Kale et al. particularly argue for *"interfaces that give [users] more freedom to express their priorities"* [12]. We focus on eliciting priorities regarding specific attributes or attribute values of items. Interfaces to explicitly specify attribute-based preferences are diverse and have long been part of dedicated visual analytics tools. For example, *LineUp* provides a visual editor that enables users to interactively specify linear and roof-like mappings of attribute values to desirability scores (including optional filters) [6]. Recent research has recognized the need for systematically investigating the space of means for preference elicitation. Based on linear attribute mappings, *WeightLifter* proposes a systematic exploration of the continuous multi-dimensional weight space underlying the creation of item rankings [13]. Schmid and Bernard surveyed different approaches for mapping numerical or categorical attribute values to desirability scores [14]. Their proposed taxonomy includes eight types of attribute scoring functions (ASFs) with different complexities. Such value functions have also been used in multi-attribute value theory to graphically elicit how the desirability changes as the attribute value changes [15]. An interface for interactively creating ASFs has been implemented in the interactive ranking creation tool *RankASco* [16] but the spectrum of function types has not been studied further. We apply the concept of attribute scoring functions for the design of different preference elicitation interfaces to study their impact on the sensitivity of metrics.

Empirical Research on Visual Decision Support – Evaluations of decision tasks mostly rely on qualitative methods or subjective ratings of how users experienced the use of a visualization to assist their decision tasks. Gettinger et al., for example, compare tables, heatmaps, and parallel coordinates based on self-reported confidence in the decision that participants made [17]. The same measure is used by Chakhchoukh to compare visual encodings for trade-off analysis [18]. While subjective metrics are useful, they are not suited to establish a rigorous comparative assessment of decision support tools.

Objective metrics that assess multi-attribute decision quality are rare. Previous works that study decisions made under uncertain outcomes [19]–[21] or with AI assistance [22] compare expected pay-offs or correctness as a single-criterion metric that builds upon decision alignment with ground truth data. In contrast, multi-attribute decisions pose the challenge of missing ground truth and conflicting criteria. Arciniegas et al. compare three map-based decision support tools based on a weighted-sum approach among decision attributes where weights have been prescribed by experts [23]. Barth et al. prescribe the preferences to be specified and use completion times and across-subject item agreement to compare an attribute-based to an item-based approach in a ranking scenario [24]. Oral et al. empirically investigated to what extent performance in judgment tasks is a sufficient predictor of performance in decision tasks [25]. Similarly, quality metrics that build upon normalization with respect to the best and worst items in the dataset have also been used to assess the effects of different personality traits on decision quality [26], [27]. However, being based on prior knowledge about the best and worst items, these metrics do not take the decision-maker's subjective preferences into account.

Some objective metrics explicitly incorporate subjective preference information. Dimara et al. have investigated three elementary visualizations' abilities to assist decision-making tasks [1]. In contrast to previous metrics, they propose an objective decision quality metric that explicitly considers the decision-maker's subjective preferences. It captures the consistency between a choice and the decision-maker's reported desirability of low or high attribute values. However, users might not be able to fully represent their preferences with a direction alone, e.g., preferences involving neither too low nor too high values.

Our work extends the body of user studies that are inspired by Dimara et al.'s decision quality metric based on self-reported preferences. Laurillau et al. propose a simplification that uses performance on the primary decision criterion as a proxy for overall decision quality [28]. Dy et al. use the metric to compare several multidimensional chart types and data volume in multi-objective decision tasks [29]. Although they observe differences in decision time, they report no strong performance advantage for any chart type in terms of decision quality. Similarly, Milutinović et al. apply the metric to study the influence of visual saliency on multi-attribute decisions using parallel coordinates and scatterplot matrices [30], again reporting largely comparable decision quality across visualization conditions. Finally, Kuhlman et al. investigate different item-based preference elicitation methods for interactive ranking [31], but the implications of such elicitation choices for assessing decision quality remain unclear.

Beyond this line of work, similar observations have been reported in related visualization decision studies. For example, Mosca et al. find that adding interaction to a visualization does not significantly improve reasoning accuracy compared to a static representation [32]. More broadly, a systematic review by Eberhard [33] concludes that visualization effects on judgment and decision-making are often small or inconsistent across studies. Taken together, these findings illustrate a recurring challenge: when decision outcomes appear similar across conditions, it is difficult to determine whether the visualizations truly perform equivalently or the employed decision quality metrics lack discriminative power. This may foster premature "no difference" conclusions, reduce the usefulness of evaluations as design feedback, and increase the risk of false negatives in comparative studies. In applied decision-support settings, such ambiguity can also translate into uncertainty about which interface to deploy for a given decision context.

Research Gap – Most comparative studies focus on determining which visualization technique best supports decision tasks. However, existing objective metrics for decision quality may inadequately capture the complexity of users' subjective preferences. In particular, there is insufficient empirical evidence on how varying the expressiveness of elicitation interfaces impacts the sensitivity of these decision quality metrics. Additionally, empirical replications designed to validate or refine previous findings on decision quality metrics in visualization contexts remain rare. Our study addresses this gap by examining how different levels of preference elicitation expressiveness affect the sensitivity of the tested decision quality metric in a controlled decision scenario. By partially replicating previous research, we also contribute empirical validation that helps strengthen methodological foundations for evaluating decision quality in visualization research.

III. Research Hypotheses

Prior works observed inconclusive responses for the decision quality metric among scatterplot matrices, radar charts, parallel coordinates, heatmaps, and tabular visualizations [1], [29], indicating a comparable performance of the visualization conditions. One explanation is a lack of metric sensitivity due to an imprecise preference elicitation: *"participants may not be able to perfectly express (or be aware of) their criteria preferences"* [1]. In other words, the differences between visualization conditions might have been too small to be captured with a decision quality metric that relies on a binary elicitation (low or high values preferred).

Following this assumption, we aim to test the metric with different preference elicitation methods as an experimental factor: What is the effect of preference elicitation interfaces with different expressiveness levels on the sensitivity of the decision quality metric, i.e., its ability to capture subtle but meaningful differences across conditions such as visualization techniques? Expressiveness refers to the complexity of functions that participants are able to specify in order to reflect which attribute values they prefer and to what extent. This relates to both the representational degrees of freedom of the interface and its practical ability to help participants articulate nuanced preferences. While these aspects are conceptually distinct, they are difficult to isolate experimentally and are therefore studied jointly. We use the term expressiveness operationally to refer to an interface's representational flexibility in terms of degrees of freedom and interaction options. This flexibility can make it easier for participants to state preferences in more detailed ways, but it should not be read as a direct measure of the nuance of their underlying preferences. With our study, we hope to contribute a first understanding of whether and how metric sensitivity can be improved by using an appropriate preference elicitation method. Prior to data collection, we framed the following research hypothesis:

H_r **Metric sensitivity increases with expressiveness** – Increasingly expressive preference elicitation interfaces may enable users to express more nuanced preferences, thereby changing how the downstream decision quality metric captures differences between conditions.

However, expressiveness needs to be weighed against cognitive load. Similar to phenomena like bounded rationality [34] and choice overload [35], people might be overwhelmed by the complexity of representations or interface usage. Contrary to our expectations, this might hinder their ability to express nuances of their internal preferences.

Metric sensitivity is likely influenced by additional factors, e.g., (in-)dependence of decision attributes or realism of participants' self-reported preferences. However, following a top-down approach, we decided not to aim at their isolation as long as we do not know whether we will observe any effect at all. Our observations might provide a starting point for an isolation of the involved influences. We assess hypothesis H_r through five sub-hypotheses. Looking at which sub-hypotheses are supported can increase our understanding of the ways in which the expressiveness of the preference elicitation relates to the sensitivity of the downstream metric for decision quality.

$\boldsymbol{H}_1$ **No difference with low expressiveness** – The choice consistency between the two visualizations does not conclusively differ when using a preference elicitation interface with low expressiveness. We expect to confirm previous results that found no conclusive difference and hypothesized that the elicitation might have been too simple to capture the full complexity of user preferences [1].

$\boldsymbol{H}_2$ **Difference with higher expressiveness** – The choice consistency between the two visualizations differs when using preference elicitation interfaces with medium and high expressiveness. We expect interfaces with medium and high expressiveness to allow participants to report their preferences more precisely, thus reducing noise in the decision quality observations and revealing more subtle differences between the visualizations.

$\boldsymbol{H}_3$ **Larger differences with higher expressiveness** – The choice consistency between the two visualizations differs more when using a preference elicitation interface with medium expressiveness than it differs when using an interface with low expressiveness. Likewise, it differs more when using a preference elicitation interface with high expressiveness than it differs when using an interface with medium expressiveness. We expect the decision quality metric based on preference elicitation with medium expressiveness to capture more subtle differences between the visualizations than the metric based on low expressiveness, because medium expressiveness comes with more flexibility for reporting one's preferences. In line with this, we expect the metric based on preference elicitation with high expressiveness to capture even more subtle differences than the metric based on medium expressiveness, because high expressiveness offers additional degrees of freedom for reporting one's preferences.

$\boldsymbol{H}_4$ **Larger consistency with higher expressiveness** – The choice consistency using a preference elicitation interface with low expressiveness differs from the choice consistency using a preference elicitation interface with medium expressiveness and high expressiveness. We expect the absolute choice consistency to be higher when using a preference elicitation interface with medium and high expressiveness than for the low expressiveness baseline condition, because we expect the mismatch between self-reported preferences and chosen decision item to be smaller in response to the increased precision of the preference elicitation.

$\boldsymbol{H}_5$ **Less variability with higher expressiveness** – The variability in responses is lower when using a preference elicitation interface with medium and high expressiveness compared to a preference elicitation interface with low expressiveness. As a result of reduced noise (cf. $\boldsymbol{H}_2$ and $\boldsymbol{H}_3$), we expect lower variability in the decision quality scores of participants using elicitation interfaces with medium and high expressiveness compared to those participants using a preference elicitation interface with low expressiveness.

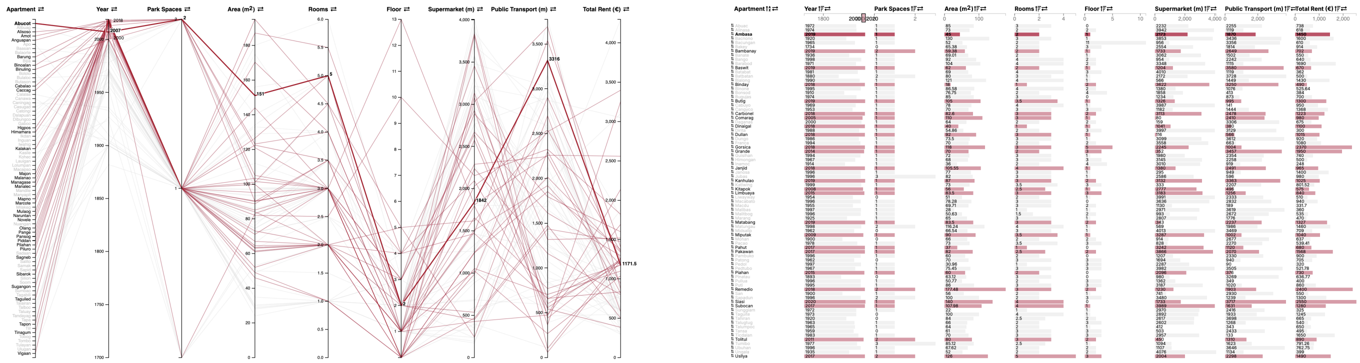
IV. Experiment

The primary goal of our experiment is to investigate how different preference elicitation methods affect the sensitivity of decision quality metrics, not to compare visualization techniques per se. We aim to explore the impact of different interfaces that allow users to explicitly specify their preferences regarding decision attribute values using attribute scoring functions [14]. Our experiment addresses the call for more flexible graphical elicitation interfaces [12] in the visualization literature (cf. Section II) as well as the call for eliciting how strongly people prefer certain options over others [15] in the behavioral economics literature. It serves the overall goal of determining decision quality as the consistency between a decision made and self-reported preferences. We compare three preference elicitation interfaces according to how well they help carve out small quality differences between decisions made with two different visualization techniques. We choose the interfaces to cover a range of representational flexibility for specifying decision preferences at different levels of complexity. In a between-subjects online experiment, we recruited 548 participants who were randomly assigned to one of the three preference elicitation interfaces. The participants were asked to solve a preference elicitation task as well as one decision task with each visualization. For the preference elicitation, they were asked to rate the importance of decision attributes and to use the preference elicitation interface to specify the desirability of the attributes' values. For the decision tasks, they used the respective visualization to interactively explore a set of options and make their decision. They received explanations about the data, visualizations, and preference elicitation first. This included a training in interacting with the preference elicitation interfaces and visualizations. After using each visualization, they briefly reflected on the decisions they made. Finally, they reflected on the preference elicitation and on which visualization they favor for decision-making.

A. Design of Preference Elicitation Interfaces

Expressing one's preferences regarding decision attributes involves both each attribute's importance as well as the desirability of its individual attribute values. We compare three experimental conditions. In all conditions, we ask participants to rate each attribute's importance for their personal decision-making on an 11-point Likert scale with 0 being *"not important at all"* and 10 being *"very important"*. We then create three different interfaces to elicit the decision-maker's preference information regarding attribute values. Decision-makers use these interfaces to specify numerical scores that reflect how (un)desirable a certain attribute value is for the decision-maker. For example, they might assign a high desirability score to a low rent value to express their preference for cheap apartments. These desirability scores are collected for all possible values of a decision attribute.

To represent preferences over attribute values, we use attribute scoring functions (ASFs) that map attribute values to numerical desirability scores in the range $[-1, 1]$ (from highly undesirable to highly desirable), allowing participants to express both favorable and unfavorable preferences. We



(a) Parallel Coordinates (PC) (b) Tabular Visualization (TV)

Fig. 2. Experiment stimuli for the decision task (*"Which apartment do you choose to rent?"*) showing the manipulated within-subjects experiment factor *visualization* with levels (a) parallel coordinates and (b) tabular visualization. Dataset of 80 apartments to choose from. Range selection applied to attribute *year* and single data item highlighted on hover.

build on the ASF taxonomy by Schmid and Bernard [14], focusing on continuous and ordinal attributes relevant for everyday decision-making.

In this experiment, we aim to explore the interplay of expressiveness and interpretability of ASF interfaces for preference elicitation. We are interested in the effect of varying levels of expressiveness on the sensitivity of the downstream decision quality metric. Does a highly expressive interface for preference elicitation drastically increase the metric's sensitivity? Or does it induce noise itself, because decision-makers get lost in the large design space of attribute scoring functions and thus settle with an approximate solution rather than striving to find a more precise representation? To find answers, we test ASF interfaces with gradually increasing degrees of freedom by increasing the number of control points and moving from linear function segments to non-linear segments. As a basis, we consider the types of functions for numerical attributes that are proposed in the ASF taxonomy [14]. We do not consider the proposed discontinuous and quantile-based types, because we believe that they are less relevant for everyday decisions like an apartment hunt. We distinguish three expressiveness levels, i.e., low, medium, and high (Figure 1). Their design is informed by a core concept of multi-attribute value theory: in addition to the *direction* of preference, e.g., many rooms in an apartment are more desirable than few rooms, humans also possess an idea of the *strength* of their preference, e.g., slight preference for a two-room apartment over a one-room apartment but strong preference for a four-room apartment over either of the previous two. Unlike the direction, the strength of preference cannot be observed by inspecting a participant's actions and needs to be elicited explicitly [15].

Low Expressiveness – The low expressiveness level captures the general direction of preference, i.e., whether the minimum or maximum is considered the most or least preferred value. Its main advantage lies in simplicity, as it can be easily represented through familiar user interface elements. In our case, participants indicate their preference direction, i.e., whether they prefer the values of an attribute to be high or low, using radio buttons (Figure 1a). This level corresponds to a two-point function with one linear segment, i.e., the two-point linear category as described in the ASF taxonomy [14]. As a drawback, it does not support the specification of the strength of preference. This condition replicates parts of a previous study [1].

Medium Expressiveness – The medium expressiveness level allows participants to specify any attribute value as the most or least preferred value, beyond just the minimum or maximum. Variations of this approach have been used in decision support tools like *LineUp* [6] and *RankASco* [16], and have also been employed to measure decision quality in comparative studies [29]. In addition to preference direction, this level allows to specify roof or ramp functions to reflect preferences that are not monotonically increasing or decreasing. For instance, ramp functions can be useful in scenarios where lower/higher attribute values are preferred only up to a saturation point. Roof functions can be helpful when preferences invert beyond such a saturation point. Participants express their preferences in a function editor by dragging the three control points of a default roof function to reflect which attribute values they prefer and to what degree. The restriction to three control points limits the possible function shapes that represent preference strength. To preserve mathematical consistency, endpoint control points can move vertically along the desirability axis only, whereas the central control point is freely adjustable. This level corresponds to a three-point function with two linear line segments (Figure 1b), an instance of the multi-point continuous category in the ASF taxonomy [14].

High Expressiveness – The high expressiveness level provides maximal flexibility, allowing participants to define virtually unrestricted mappings between attribute values and preference scores. Both direction and strength of preference can be specified in detail. This approach was used in the ranking creation tool *RankASco* [16]. While this level can best represent complex preference structures, it poses a higher risk of user misinterpretation. Participants sketch their preference function by adding control points that are interpolated via a spline curve. Unlike the *RankASco* interface [16], we start from a blank canvas to avoid biasing users towards any function shape. Participants can freely add, move, or remove intermediate control points to express their degree of preference for certain attribute ranges. As with medium expressiveness, control points at attribute range endpoints can only move vertically. This level corresponds to a multi-point function with any number of non-linear line segments (Figure 1c), another instance of the multi-point continuous category [14].

B. Visualization Techniques

In addition to the preference elicitation interface, we also manipulate the visualization that participants use to make their decision to investigate the metric's ability to detect differences between conditions. There are several elementary visual encodings that have been widely examined in decision-making research [1], [29], [30]. To limit the time required from each participant while still enabling a meaningful comparison, we selected two encodings. Parallel coordinates (PC) and tabular visualization (TV) represent distinct approaches and showed contrast in prior work, although performances were largely comparable among common chart types [1], [29]. Because PC and TV produced limited baseline differences, this design provides a challenging test case for examining how the metric behaves under different elicitation interfaces.

Parallel coordinates are used in a standard layout, i.e., attributes are represented as parallel vertical axes and decision items are depicted as polylines that intersect the axes at their corresponding values (Figure 2a). The tabular visualization encodes cell values by bar length (Figure 2b). Interactions include highlighting of individual decision items via hovering, dynamic filtering via range selection on one or multiple attributes, and attribute reordering via drag and drop. The tabular visualization also allows for manual reordering of decision items as well as automatic sorting by a specific attribute.

C. Study Design

The experiment followed a 2×3 mixed design. The *visualization* (parallel coordinates, tabular visualization) was a within-subjects independent factor, such that a participant was exposed to both visualizations. The *preference elicitation* (low, medium, high expressiveness) was a between-subjects independent factor, such that participant report their preferences with an interface of either expressiveness. We randomly assigned participants to the between-subjects condition.

After a training phase that covered the respective preference elicitation interface and the two interactive visualizations, participants reported their preferences for all decision attributes according to their assigned condition. Each participant then performed two decision tasks: one with a tabular visualization and one with parallel coordinates. To avoid that participants recall their prior decision in the second task, we used two different datasets with identical semantics to pair with the visualization. To avoid order effects for the within-subjects factor, we implemented a complete counterbalancing for the visualization presentation order as well as the visualization/dataset pairing.

D. Data, Tasks, and Procedure

To tap into the natural decision-making behavior of participants, they need to identify themselves with the given

Apartment	Year	Park Spaces	Rent (€)	Area (m^2)	Rooms	Floor	Next Supermarket (m)	Next Public Transport (m)
Alisoso	2013	1	1520	144.02	3	3	3454	1318
Banate	1937	1	1225	90	3	2	3519	3005
Salito	1900	1	1792	112	3	1	1491	2977
Managase	2017	2	1100	105.01	3	0	1055	704
Balaorte	1998	1	780	76	3.5	3	3564	388

Fig. 3. Examples from the dataset participants used for the decision task.

decision problems. The narratives used in the study thus need to reflect familiar decision-making situations, i.e., from the participants' lives, as closely as possible. We used two different *decision scenarios* that the participants have likely experienced themselves at some point in their lives: holiday accommodation and apartment hunting.

For the training of the visualizations and preference elicitation, we used the same decision scenario and dataset as Dimara et al. [1]. Participants were asked to imagine that they were planning a holiday and looking for the best hotel for their stay. The hotels were defined across eight attributes: price per person, hotel quality, archaeological offer, landscape, night life, security level, sport activities, and child friendliness. To convey how the visualizations encoded the data, participants were presented with a table showing a subset of five hotels across four attributes next to parallel coordinates and a tabular visualization depicting the same subset. Annotations explained the fundamental visual mapping.

Participants who were assigned a medium or high expressiveness level then received preference elicitation training. After textual explanations of how to define and interpret a preference function (Figures 1b and 1c), they were invited to try out the interactive interface themselves to specify preferences regarding hotel quality as a predefined attribute. To assess whether participants understood the explanations, the training concluded with a *preference elicitation training task* that asked participants to select one out of four depicted functions that best reflected a given textual preference description.

All participants then received visualization training on the same dataset. This particularly focused on the interactive capabilities: item highlighting, range selection, and attribute reordering for both visualizations as well as item reordering and attribute sorting for the tabular visualization. For each interaction, we provided a textual explanation alongside an interactive visualization of five hotels for participants to directly try out the interaction themselves with interactive feedback confirming task completion. To assess whether participants had understood how to work with and interpret both interactive visualizations, each visualization training concluded with three *visualization training tasks* that participants were asked to solve with a visualization of 80 hotels. To this end, we used simple analytical tasks [36]: naming the item that has a certain attribute value and finding the value of another of its attributes (value retrieval task) as well as counting how many items have a certain attribute in a given value range (range task).

For the actual decision-making that followed the training, we reused an apartment hunting scenario and dataset [37] from a previous study on ranking creation [24]. Participants were asked to imagine that they were about to move to a different city and searching for the ideal apartment to rent.

To maintain comparability with attribute types and counts used in similar studies, we modified the data by removing non-numerical attributes and synthetically generating two numerical attributes (randomly generated attribute values in a semantically meaningful range). The synthetic attributes were chosen to provoke diverse plausible forms of preferences. In the end, the apartments were defined across eight attributes: rent, living area, number of rooms, floor, construction year, number of parking spaces, distance to closest supermarket, and distance to closest public transport. Again, five example apartments were depicted in a standard table (Figure 3).

This introduction was followed by the *preference elicitation task*. All participants filled in Likert scales to indicate how important they considered each of the above attributes for their upcoming decision. For each attribute with a rating other than *"not important at all"*, the participants then reported the desirability of the attribute's values as described in Section IV-A. For the low expressiveness condition, participants used radio buttons to report whether they prefer high or low values of an attribute. For the other two conditions, they interactively specified a three-point linear or a multi-point non-linear function that maps the attribute's values to desirability scores in the range $[-1,1]$. In contrast to Dimara et al., where preference elicitation was performed at four different time points, we elicited our participants' preferences only once before presenting the decision tasks in order to keep the study duration at a manageable level [1]. We consider the risk of a priming effect negligible, as their results showed only marginal variation between desirability scores computed on preferences elicited before and after a decision task (interquartile range of 0.0045 for scores in $[0,1]$). Any priming would also affect all experimental conditions equally, such that we do not expect a bias regarding the comparisons of mean differences on which our conclusions are based.

Each participant then performed two *decision tasks*, one with each visualization. They were prompted to use the visualizations to interactively explore different sets of 80 apartments and choose their most preferred apartment among each set. Participants made their choice by copying the apartment's name in a text field above the visualization. After each decision task, participants were asked to evaluate the choice they just made. After completion of both decision tasks, they reported which of the two visualizations they favor and why.

As we aimed for a much larger number of participants and refined research hypotheses compared to previous studies [1], [29], our experiment differed in some aspects. We tested two additional preference elicitation conditions, effectively turning the study into a mixed design with preference elicitation as a three-level between-subjects factor. Our targeted sample size was only feasible in a crowdsourced setting. The training therefore took the form of an indirect online tutorial rather than in-person explanations. Two post-task attention check items, one instructional manipulation check [38] and one nonsensical item [39], helped screen out inattentive participants. We also augmented the setup with additional control variables that we deemed important for interpretation of our observations but that were not focal variables of the study. At the end of the study, we asked for ratings regarding the perceived cognitive load [40] associated with the preference elicitation and regarding the participants' perception of how accurately they could express their preferences. As part of the demographic questions, we also asked participants to rate their personal level of decisiveness and their experience with using or developing visualizations of data. Finally, to complement the self-report of visualization experience, we conducted the Mini-VLAT [41] at the very beginning of the questionnaire. To summarize, our procedure consisted of five phases: Mini-VLAT, training, preference elicitation, decision-making (including assessment), and post-task survey. The questionnaire with detailed experiment instructions is available in the supplemental materials.

E. Participants

We conducted an a priori power analysis to calculate the minimum sample size of $N = 600$. From the observations of the previous study [1], we computed an effect size of $d = 0.2$. On this basis, we performed a power analysis for a two-tailed paired samples t-test using G*Power [42]. It resulted in a minimum sample size of $N > 199$ for the preference elicitation condition with low expressiveness that replicates the previous study. Our goal was to obtain 0.8 power to detect the effect size of $d = 0.2$ at the standard 0.05 alpha error probability (following conventions in visualization and human computer interaction [43]). To have an equal number of participants across all three levels of the between-subjects factor, we need the same number of participants also for the two remaining preference elicitation conditions. Thus, data of $N = 600$ participants was collected. The calculation details are provided in the supplemental materials.

As we aim to study decisions made by all kinds of people, we do not impose restrictions regarding our target population, particularly not regarding prior experience in visualization. Participants were recruited through the crowdsourcing platform Prolific[1] and paid 8,50€/h for participation. They could only participate once. We required them to be fluent in English, between 18 and 80 years old, and to have a 95% approval rate for their previous submissions on the platform. Prolific invited participants from the diversity of more than 186 000 eligible workers at the time of execution. Recruiting continued until the target sample size was reached with complete submissions. Withdrawn, timed out, and rejected submissions were not counted. Participants completing the study with two failed attention checks, missing consent, or in less than six minutes (i.e., the maximum duration of the Mini-VLAT [41]) were given the opportunity to withdraw their submission to avoid rejection penalties. With 42:23 minutes, the median study duration was close to our prior estimate of 40 minutes.

To verify our technical implementation, received responses, and the estimated completion time, we piloted our setup on a small number of participants. Once the identified issues were solved, we distributed the study to the remaining number of participants. As Prolific made sure to resample objectively invalid submissions from the sample, we expected the subsequent drop-out rate to be fairly low. To balance a fair payment

[1] www.prolific.com

of workers with our financial constraints, we decided to not over-recruit, being aware that unexpected drop-outs might decrease the effect size of observed outcomes.

F. Metrics

Accuracy is the most important outcome measure to study the impact of preference elicitation on assessing the quality of decisions made with parallel coordinates and tabular visualizations. Additional subjective metrics referring to the decision task include the choice assessment and technique preference. To provide further context for our research focus on the impact of preference elicitation, we augment the subjective metrics with perceived cognitive load and precision referring to the preference elicitation task.

Choice Consistency – There is no commonly agreed-upon approach to objectively assessing the accuracy of a decision in a multi-attribute setting. We join previous studies [29], [30] in deploying the principle suggested by Dimara et al. [1], who observe the consistency between a chosen item and the decision-maker's self-reported preferences as an indicative measure of accuracy. We intentionally retain this scalar metric because our goal is to study how preference elicitation affects the sensitivity of the downstream decision quality metric rather than to analyze the full structure of elicited preference functions. However, we refer to it as choice consistency to emphasize that the metric does not imply any objective correctness. Each decision item receives a desirability score that reflects how well it matches the decision-maker's self-reported preferences as expressed through the elicitation interface. The scores are normalized to the range $[0, 1]$, with 1 being a fully consistent choice with respect to the self-reported preferences. We apply the previous weighted-sum approach to compute each item's desirability score. For the low expressiveness condition, the approach can be left unchanged, i.e., each attribute's value is normalized and inverted if dictated by the preference direction. For the two additional conditions, only a subtle modification is necessary. The specified preference functions already map the attribute values to preference scores, such that we read an attribute value's score directly from the function. These values are then multiplied with the respective attribute's reported importance and summed up to the item's desirability score. For all conditions, we normalize the desirability scores across all items.

Choice Assessment – After each decision task, participants reflected on the choice they just made regarding satisfaction, confidence, easiness, and attachment. All responses were reported on an 11-point Likert scale, with anchors adjusted according to the nature of each question.

Visualization Preference – After all decision tasks had been completed, the participants reported their overall visualization preference, again on an 11-point Likert scale. The participants were asked how helpful they found each visualization for making their choice, with 0 being *"not at all helpful"* and 10 being *"very helpful"*. They were also invited to explain their ratings in a text field.

Control Variables – As control variables related to the preference elicitation, we identified the perceived cognitive load associated with the elicitation process and the perceived precision of the reported preferences. We further considered the participants' general decisiveness, their visualization literacy, and demographics like gender, age, and education level.

We included two subjective measures that capture the cognitive load and perceived precision associated with the preference elicitation on an 11-point Likert scale. For cognitive load, we used a widespread rating scale from cognitive load theory research [40] that asks participants how much mental effort they have invested, in our case in specifying their preferences. For the perceived precision of the preference elicitation, we adapted the wording of the NASA-TLX [44] item regarding task performance and asked participants how accurately they could express their preferences.

Finally, participants rated their personal level of decisiveness on a 5-point Likert scale. Our assessment of visualization literacy was two-fold. On the one hand, participants rated their experience with using or developing data visualizations on a 5-point Likert scale. We complemented this subjective self-report with an objective literacy score that we obtained by conducting the Mini-VLAT [41] at the beginning of the questionnaire.

G. Implementation

We set up and hosted the questionnaire using the platform SoSci Survey[2]. All questions were marked as mandatory except for participants' free-text justification of their technique preference rating. All interactive components (i.e., parallel coordinates and tabular visualization as well as preference elicitation interfaces with medium and high expressiveness) were implemented based on Next.js[3] and D3.js [45] and displayed as part of the questionnaire. The implementation of the preference elicitation interfaces with medium and high expressiveness is based on the *MultiPointContinuous* component of *RankASco* [16] licensed under MIT.

While preserving its aspect ratio, each visualization stretched across the entire horizontal display space as well as the remaining vertical display space below the question item asking for the participant's decision. Each visualization could accommodate the eight attributes and 80 decision items without the need for scrolling in either direction, and with legible fonts for display resolutions of 1920×1080 and higher. In the beginning, we explicitly asked participants to maximize the browser window that displays the study.

H. Data Exclusion

From 600 submissions, we screened out 15 submissions based on the preregistered exclusion criteria: incomplete submissions, two of two failed attention check items, and extremely high completion times (i.e., more than three standard deviations away from the mean).

In addition to the preregistered criteria, we needed to screen out 37 submissions with missing or invalid preference specifications, which could not be validated at the time of data collection due to technical limitations of the platform. This applies to 15 submissions where attributes were rated as relevant

[2]https://www.soscisurvey.de
[3]https://nextjs.org

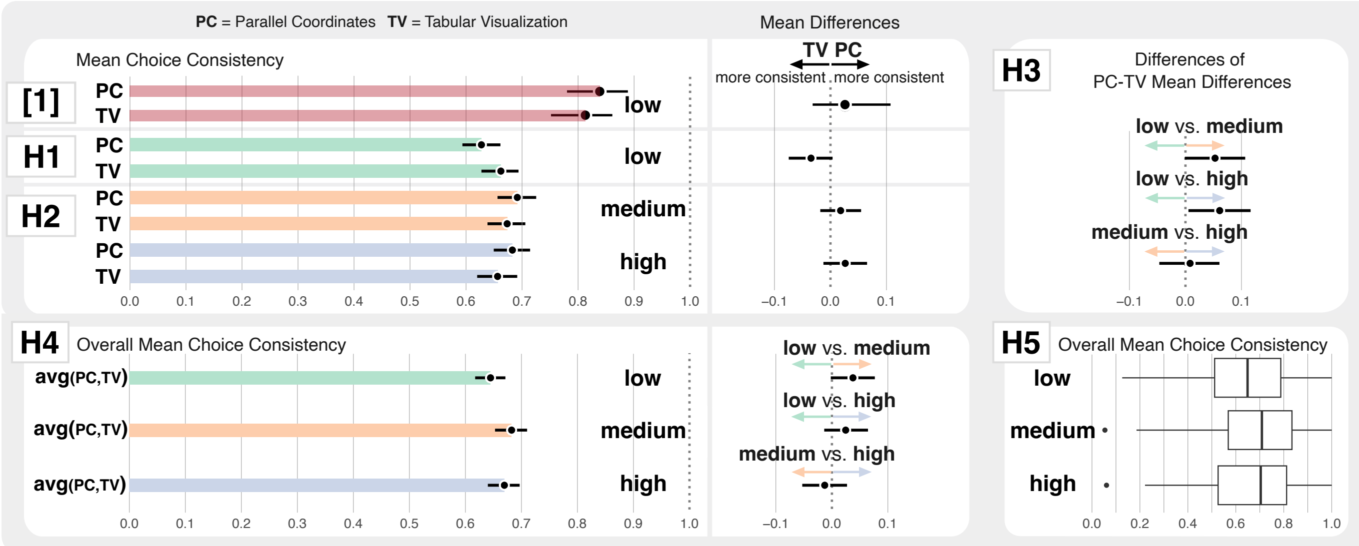


Fig. 4. 95% confidence intervals show mean choice consistency per condition (left), mean differences between conditions (center), and differences of PC-TV differences between expressiveness levels (top right). While the suggested slight advantage for TV with low expressiveness (H1) deviates from previous findings, our study generally replicates the inconclusive difference between PC and TV that was reported before ([1], H1, H2). Medium expressiveness showed the highest overall choice consistency (H4) and the numerically lowest variation as a representative of noise (H5), supporting weak evidence of a plateau effect: PC-TV differences peaked at medium expressiveness, with further increases in expressiveness leading to little additional sensitivity benefits (H3).

for the decision but had no preference specification, where participants left all default functions untouched, or where a participant defined an ambiguous three-point linear function. It also applies to another 22 submissions where participants selected uniform importance ratings across all attributes. This is called *straightlining* and generally considered a sign of careless responding [46].

V. Results

In total, we had 548 participants (n=188 for low, n=174 for medium, and n=186 for high preference elicitation expressiveness). The conditions are slightly unbalanced due to natural variation from post-screening drop-outs. The participants were aged under 20 to over 60 with 72.8% aged between 20 and 39 years. 236 of them identified as female (43.1%), 309 identified as male (56.4%), and 3 identified as diverse (0.5%).

Following the view that evidence lies on a continuum rather than being dichotomized by significance cutoffs, we base our inferences on interval estimations of effect sizes that we report graphically [47], [48]. All confidence intervals (CIs) are 95% BCa bootstrap confidence intervals [49]. Aligning with the previous study [1], this approach also supports comparability for replication.

A. Planned Confirmatory Analyses

All analyses in this section were planned and preregistered[4] prior to data collection. Each subsection corresponds to one of our five sub-hypotheses (Section III). Differences between visualizations are computed on a per-participant basis, while differences between preference elicitation conditions are computed between-subjects.

[4]https://doi.org/10.17605/OSF.IO/KW8UJ

With mean consistencies between 0.63 and 0.69 across all conditions, participants rarely selected the item that exhibited the best match with their self-reported preferences. Although our observed differences between conditions are generally small and do not provide clear evidence for systematic effects, we noticed some interesting phenomena regarding the interplay of preference elicitation and the sensitivity of the decision quality metric, which we report and discuss in the following.

H_1 **No difference with low expressiveness** – We compare the mean consistency scores of tabular visualization and parallel coordinates for the low expressiveness condition (green bars in Figure 4, H1, left). The confidence interval showing the mean differences in consistency between the two techniques barely overlaps at zero (black CI in Figure 4, H1, right, CI $[-0.075, 0.004]$). While the point estimate suggests a slight advantage of TV, the inclusion of zero does not provide conclusive evidence for a difference in consistency between the two visualizations when using the preference elicitation interface with low expressiveness.

H_2 **Difference with higher expressiveness** – Like for H_1, we compare the mean consistency scores of the two visualizations, this time for the medium and high expressiveness conditions (orange and blue bars in Figure 4, H2, left). Both interval estimates for the mean differences clearly include zero (black CIs in Figure 4, H2, right). The results thus do not provide conclusive evidence for a difference in consistency between the two visualizations when using the preference elicitation interface with medium or high expressiveness.

H_3 **Larger differences with higher expressiveness** – We compare the difference in PC-TV differences between all three expressiveness levels. Revisiting the individual PC-TV differences observed for H_1 and H_2, the interval estimate for low expressiveness (black CI in Figure 4, H1, right) only marginally overlaps with the interval estimates for medium and high expressiveness (black CIs in Figure 4, H2, right). Note that the point estimates suggest a change in the sign of the differences, from TV being more consistent on average in $\boldsymbol{H}_1$ to PC being more consistent on average in $\boldsymbol{H}_2$. This reversal of direction means that the signed difference-in-differences becomes positive even though the absolute PC-TV differences are numerically larger under low expressiveness. Consequently, the interval estimate for the difference of differences barely overlaps at zero when comparing low to medium expressiveness (top black CI in Figure 4, H3, CI $[-0.001, 0.107]$) and does not include zero when comparing low to high expressiveness (middle black CI in Figure 4, H3, CI $[0.005, 0.116]$). This may indicate that the choice consistency between the two visualizations differs more when using a preference elicitation interface with medium or high expressiveness than it differs when using an interface with low expressiveness. However, the evidence is weak and should be interpreted with caution.

We further observe that there is a considerable overlap between the interval estimates for the PC-TV differences at medium and high expressiveness (black CIs in Figure 4, H2, right). Consistently, the interval estimate for the difference of differences between medium and high expressiveness is almost centered at zero (bottom black CI in Figure 4, H3). This indicates that we have no evidence for a difference between medium and high expressiveness in terms of the magnitude of consistency differences captured between the visualizations.

$\boldsymbol{H}_4$ Larger consistency with higher expressiveness – We compare the overall mean consistency scores independent of the visualization across the three expressiveness levels of the preference elicitation. We measured the highest mean consistency with the metric that was based on the preference elicitation with medium expressiveness (orange bar in Figure 4, H4, left). The interval estimate for the mean consistency differences between low and medium expressiveness barely overlaps at zero (top black CI in Figure 4, H4, right CI $[-0.002, 0.077]$). While the point estimate suggests that the metric based on medium expressiveness measures higher choice consistency than the metric based on low expressiveness, the inclusion of zero does not provide definite evidence. The evidence is even weaker for a difference between the metrics based on high and low expressiveness (middle black CI in Figure 4, H4, right, CI $[-0.014, 0.065]$). The interval estimate for the mean differences between medium and high expressiveness is almost centered at zero (bottom black CI in Figure 4, H4, right). This indicates that we have no evidence for a difference in measured consistency scores between medium and high expressiveness.

$\boldsymbol{H}_5$ Less variability with higher expressiveness – We computed box plots to summarize the variation of responses for each preference elicitation expressiveness level (Figure 4, H5). We compare the interquartile ranges, i.e., the spread of the middle 50% of responses. Variability is numerically lowest in the medium expressiveness condition (interquartile ranges: 0.276 for low, 0.266 for medium, and 0.286 for high expressiveness). However, the differences between conditions are small and do not provide clear evidence for a systematic change in response variability due to varying expressiveness. An ANOVA test can be found in the supplemental materials.

Subjective Choice Assessment and Technique Preference – Participants rated four different choice assessment metrics when asked to evaluate the choice they just made. We found no evidence of a difference between decisions made with PC versus TV for choice satisfaction, confidence, and attachment. For perceived ease, participants in the high expressiveness condition rated decisions made with TV slightly easier than those with PC. However, this difference was small and weaker in the low and medium expressiveness conditions. When asked how helpful the participants found each visualization for making their choice, our results suggest across all preference elicitation conditions that participants prefer TV over PC. We provide interval estimates of mean ratings and mean differences for each choice assessment metric and visualization preference in the supplemental materials.

B. Additional Exploratory Analyses

We report additional exploratory analyses to better understand in which respects the three preference elicitation interfaces with different expressiveness levels differ.

Perceived Precision and Cognitive Load – We did not observe differences when comparing the perceived precision and cognitive load induced by the three expressiveness levels. On a scale of [0,10], the average perceived precision was 7.65 (SD = 1.87) for the low expressiveness, 7.83 (SD = 1.85) for the medium expressiveness, and 7.77 (SD = 1.80) for the high expressiveness. The average perceived cognitive load was 7.65 (SD = 2.13) for the low expressiveness, 7.98 (SD = 2.03) for the medium expressiveness, and 7.98 (SD = 2.13) for the high expressiveness. While perceived precision and cognitive load are numerically lowest in the low expressiveness condition, the differences between conditions are small and do not provide clear evidence for a systematic effect. The respective boxplots can be found in the supplemental materials.

Complexity of Elicited Preferences – We wanted to know whether participants had taken advantage of the virtually unrestricted flexibility of the preference elicitation interfaces with high expressiveness at all. If not, this might explain negligible differences in consistency scores. As an approximation of function complexity, we inspected how many control points participants in the high expressiveness condition (n = 186) used to define their preference mappings. Across all eight decision attributes, we could observe right-skewed distributions, with a peak at four to five control points and the tails extending to 11 to 18 control points. While this indicates that people indeed defined more complex functions than possible in the other conditions, it is difficult to draw firmer conclusions without knowing their actual function shapes.

We therefore took a closer look at examples of function definitions. For a first impression, we particularly selected cases where 1) few control points were used for attributes where we expected more nuanced preferences to be plausible (like construction year) and where 2) many control points were used for attributes where minimization or maximization preferences are plausible (like rent or floor). When looking

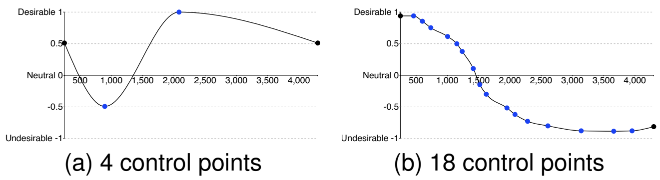


Fig. 5. We observed different strategies that participants in the high expressiveness group used to specify their preferences for *rent*.

at a function with a common number of control points, we observe that participants used few control points to reflect quite complex preferences (Figure 5a). We also observed multiple cases where a two-digit number of control points had been used to define a rather simple function, where the local effect of control points remains rather small (Figure 5b). This might indicate that some participants might not have understood whether and how they could have represented their preferences with fewer control points.

Thus, while we see initial indications of people taking advantage of the high expressiveness, more investigation is needed for a thorough overview of the extent to which participants make use of a preference elicitation interface's expressiveness. Our huge collection of more than 2700 defined preference functions from medium and high expressiveness conditions (made available in the supplemental materials) can contribute interesting findings regarding the interplay of the flexibility that an interface allows and the flexibility that people actually make use of.

Repeated Analyses with Filtered Participant Subset – We wanted to know whether our results involved statistical noise that originates from participants not knowing how to work with the visualizations or preference elicitation interfaces. We therefore repeated the reported planned analyses with only those 305 participants who passed both attention checks, the preference elicitation training task (where applicable), and at least two of three training tasks for each visualization. While the overall mean consistency scores were a little higher (0.65 to 0.75 across all tested conditions), the previously observed PC-TV differences and differences of PC-TV differences were generally reproduced.

Here, we only report on those results where the CIs showed slightly different significance levels – the full results can be found in the supplemental materials. The interval estimates for the differences of PC-TV differences ($\boldsymbol{H}_3$) and differences of overall consistency scores ($\boldsymbol{H}_4$) between low and medium expressiveness no longer include zero, strengthening our confidence in the previous findings that sensitivity and measured overall consistency seem to peak at medium expressiveness. Furthermore, the CI for overall mean consistency scores ($\boldsymbol{H}_4$) for high expressiveness has moved towards an overlap with that for low expressiveness rather than that for medium expressiveness, making the already previously observed peak at medium expressiveness stand out slightly more and indicating a trend towards similar performance of low and high expressiveness. The results also suggest that medium expressiveness leads to the lowest variability in choice consistency ($\boldsymbol{H}_5$), with low and high expressiveness performing similarly.

Repeated Analyses Grouped by Mini-VLAT Scores – To additionally control for visualization literacy as a potential explanation for statistical noise, we also repeated the analysis with participants grouped by their Mini-VLAT scores into low (0 to 4 with n=138), medium (5 to 7 with n=277), and high (8 to 12 with n=133) scores. We provide the full results in the supplemental materials. Overall, the results across all conditions show similar trends. The slight consistency advantage for TV in the low expressiveness condition ($\boldsymbol{H}_1$) seems to manifest particularly among participants with medium and high Mini-VLAT scores. The observation that choice consistency between PC and TV differs more with medium or high expressiveness ($\boldsymbol{H}_3$) seems to particularly manifest among participants with medium Mini-VLAT scores. The overall mean consistency scores ($\boldsymbol{H}_4$) are higher with increasing Mini-VLAT scores, while the differences in measured consistency between preference elicitation conditions remain inconclusive. As before, the variability in choice consistency is lowest with medium expressiveness ($\boldsymbol{H}_5$) among participants with low and high Mini-VLAT scores. Only among participants with medium Mini-VLAT scores, we cannot observe clear differences in the variability between expressiveness levels.

C. Discussion

This study set out to investigate whether increased expressiveness in preference elicitation interfaces enhances the sensitivity of decision quality metrics. Across multiple analyses, our results suggest that medium expressiveness provides clearer and more consistent effects, indicating a possible plateau beyond which further increases in expressiveness may not provide additional benefits. Below, we revisit our hypotheses, reflecting on design trade-offs involved in evaluating metric sensitivity and broader implications for preference elicitation and decision support.

Replication and Reflection on Experimental Design – Dimara et al. reported no conclusive difference between PC and TV regarding decision accuracy under low expressiveness conditions [1]. Although participants expressed a slight subjective preference for TV, the accuracy metric itself revealed no clear advantage for either visualization. Given their modest sample size (21 participants), it was plausible to attribute the inconclusive results to statistical noise. Our replication, however, utilized a significantly larger sample (548 participants), a different decision scenario, and a more challenging online setup, yet it yielded remarkably similar results.

In line with their study [1], we found no conclusive evidence for a difference in choice consistency between the two visualizations under low expressiveness ($\boldsymbol{H}_1$). Again, participants expressed a slight subjective preference for TV for making their choice. This strengthens confidence that the initial findings were not merely a small-sample artifact. In absolute terms, our consistency scores were lower compared to the previous study (red bars in Figure 4, [1], left), which might hint at a more difficult decision task. At the same time, they were comparable to the level measured in another study with a similar decision quality metric [30]. While we observed a slight trend towards lower consistency with PC compared to TV, the effect is small and likely negligible, particularly given our large sample size. Still, the deviation from the previous findings, where PC showed a marginal, non-significant advantage (black CI in Figure 4, [1], right), may be noteworthy.

Participants did appear slightly less consistent with TV under medium and high expressiveness, consistent with the direction of our expectations ($\boldsymbol{H}_2$). However, the point estimates for these differences are small, in fact smaller than we expected, and the confidence intervals include zero. Thus, we cannot draw conclusive evidence for an effect at this stage. While the slight advantage of PC under medium and high expressiveness warrants further investigation, its practical significance remains unclear and should be interpreted with caution.

Overall, our results replicate the previous findings [1] with a significantly larger sample size, observing only minimal differences in consistency between visualizations across all expressiveness levels. Thus, while confirming the robustness of the previous result [1], our findings suggest that PC and TV, given their consistently similar performance on the tested decision task, may not represent ideal test cases for evaluating the sensitivity of decision quality metrics. While studies aiming to identify a superior visualization typically compare similarly well-performing techniques (e.g., incremental improvements over state-of-the-art), a different experiment design might be required to assess a metric's ability to detect small differences between conditions. Such studies should therefore consider testing visualizations, scenarios, or tasks that can be expected to exhibit clearer baseline differences in order to determine metric sensitivity as the capture of distinctly varying outcomes. Nevertheless, such subtle baseline differences allow us to probe whether the metric is capable of detecting small effects at all.

Plateau Effect Regarding Metric Sensitivity – Our analysis of the PC-TV differences revealed important observations, even though we could not draw on substantial baseline differences: increasing the expressiveness of the preference elicitation did not lead to consistently larger differences between the visualization conditions ($\boldsymbol{H}_3$). Contrary to our initial expectations, we observed evidence resembling a plateau effect: high expressiveness does not clearly increase the differences captured by the consistency metric beyond medium expressiveness under our conditions. This finding aligns with the results for overall choice consistency ($\boldsymbol{H}_4$), which was highest at medium expressiveness and comparably lower at low and high expressiveness. The role of medium expressiveness was also reinforced by our results for the variability in choice consistency responses as a representative of noise ($\boldsymbol{H}_5$). In our exploratory analysis on presumably more experienced participants, variability was lowest at medium expressiveness and comparably higher at low and high expressiveness, although the main analysis only supported this trend numerically.

To conclude, within the scope of this study, we found no clear evidence that increasing expressiveness beyond the medium condition improved the sensitivity of the tested choice consistency metric. One possible explanation is that highly expressive preference elicitation introduces complexity without necessarily enhancing clarity. It could be that medium expressiveness provides enough flexibility to capture nuanced preferences without overwhelming participants. Cognitive mechanisms similar to those described by the well-known choice overload, in which excessive options can lead to decreased decision quality and satisfaction, may be at play here [35], [50]. Alternatively, decision theory suggests that people often do not possess fully developed preferences before encountering concrete options [51], [52]. Thus, increased expressiveness might not be beneficial if users themselves lack clarity regarding their own preferences.

Interestingly, self-reported cognitive load and perceived precision showed no meaningful differences across expressiveness levels. This could suggest that participants were either unaware of the increased cognitive demands or that their main challenge was in identifying their preferences rather than in articulating them precisely. Future research might therefore benefit from focusing on methods that support users in clarifying their preferences, rather than solely providing greater expressiveness in the interface design.

While the broader methodological question of how elicitation expressiveness affects metric sensitivity is applicable to preference-driven decision evaluations more generally, the present findings should be interpreted as evidence from one controlled decision scenario and visualization pair. They suggest that there is a balance to be struck between expressiveness and interpretability in preference elicitation. This opens up new research questions and the need for further investigation to isolate the mechanisms driving the observed effect and investigate whether such a 'sweet spot' exists across tasks, visualization types, datasets, and attribute structures. The challenge is to keep decision quality metrics both sensitive and practically meaningful, which might also include considerations of risks like overfitting.

VI. Research Opportunities and Implications

Our findings point to several promising directions for future visualization research, particularly regarding how subjective preferences are formed, elicited, interpreted, and used in decision support.

Even though the lack of ground truth makes it challenging to control for the ultimate realism of elicited preferences, our indications of a plateau effect rather than the expected continuous improvement of metric sensitivity hint at more complex mechanisms being attached to working with decision preferences than we expected. This might have implications for the development of decision support tools but also for comparative decision quality metrics that involve subjective preferences. Our results support ongoing efforts in visualization research that aim to help users explore how modifying preferences affects rankings and outcomes [6], [13], [16]. However, much of the existing work still assumes that users either know their preferences or will become aware of them during the interaction. While robust elicitation methods are important, we advocate for increased attention to preference awareness, i.e., how users come to recognize and articulate what matters to them in a given context. An open question is how the very means of eliciting preferences might shape how people formulate them. Concepts like nudging [53] suggest that such influence is possible, but future studies are needed to disentangle the mechanisms influencing preference reports in the context of visual decision support. Given the lack of ground truth to compare against, the accuracy of self-reported preferences might be approximated as the agreement of reports generated through verbalization and different graphical interfaces. Approaches like the Critical Decision Method [54] might help contextualize collected preference reports to assess their realism and maturity level. Usability issues might additionally distort preference reports and should be avoided (e.g., by providing explanation, training, and visual feedback) and controlled for (e.g., by assessing them using usability heuristics [55] or correctness of exercises).

Our results also motivate further research on when to use which level of preference elicitation expressiveness for a comparative metric. Future studies with clearer baseline differences could help refine the characteristics of the plateau effect we observed. Research might also investigate a more granular spectrum of expressiveness levels, or even adaptive approaches that allow users to adjust complexity based on decision criticality, data characteristics, or personal comfort [56]. Expanding on our attribute-based approach, this might also include modeling preferences from repeated choices between two decision items [57], [58]. Such ambitions can help identify under what conditions different levels of expressiveness lead to meaningful differences in decision outcomes. To investigate such effects more rigorously, empirical visualization research should engage with decision theory and behavioral economics, which offer rich perspectives on how preferences are formed, expressed, and influenced.

In a broader societal context, our study highlights the importance of understanding how preferences are formed, elicited, and evaluated in environments increasingly shaped by algorithmic recommendations. Recommendation systems often provide options without requiring explicit input, which may encourage users to accept suggestions passively. If so, visualization tools that encourage reflection and support the articulation of individual preferences could play a key role in reinforcing users' decision-making agency.

Given our initial evidence of the existence of a plateau phenomenon but rather small effects in general, more research is needed on the sensitivity of decision quality metrics. This includes exploring more divergent visual conditions, decision scenarios, and metric variants to better understand when and how subtle effects can be reliably detected. While it makes sense to start with elementary visualizations to develop an understanding of basic visual encodings and interactions, it would be interesting to apply similar experiment designs for comparing extensions like provenance for PC or weighted-sum ranking for TV or even for comparing complete systems like *LineUp* [6] or *PAVED* [9].

Lastly, our dataset offers several avenues for further research. The 548 participants explained their visualization preferences in free-text responses and provided over 2700 preference functions. While manual text analysis becomes difficult at this scale, large language models have become powerful tools for classifying, summarizing, and analyzing this data, e.g., by identifying recurring themes and conducting sentiment or emotion analysis. Similarly, a dedicated analysis of the collected function shapes – rather than collapsing them into a scalar consistency measure – may deepen our understanding of the nuances, with which people approach the task of expressing their mental model of preferences explicitly. We initiated a qualitative analysis of the functions by developing a dashboard[5] for their open-ended exploration. It provided insights into common articulation strategies and the extent to which people engage with available flexibility or fall back on simple heuristics that generally support our observations of choice consistency [59].

VII. Conclusions

To the best of our knowledge, we conducted the first empirical study that investigated the interplay between preference elicitation interfaces and measured decision quality by examining how different elicitation conditions affect the detection of small differences in the tested choice consistency metric. In a preregistered online experiment on decision-making with 548 participants, we partially replicated a prior study [1] and extended it with two additional levels of expressiveness in preference elicitation. Our findings confirm that commonly used visualizations, parallel coordinates and tabular visualizations, seem to yield comparable choice consistency under low expressiveness conditions, reaffirming the robustness of earlier results. At the same time, medium expressiveness showed suggestive evidence of enhanced metric sensitivity and lower variability, but further increases in expressiveness did not provide clear additional benefits. Although the overall effects observed in this study are subtle, they suggest a nuanced interplay between interface design and evaluative sensitivity in this experimental context of subjective, preference-driven tasks. Future work can build on these foundations to explore adaptive or personalized elicitation strategies, investigate richer visualization systems, and develop more robust, context-aware metrics.

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